Contribution Report: Student Management System

Course: ADVANCED WEB APPLICATION DEVELOPMENT (BUC3131)

Coursework Two

Group project: Laravel Student Management System

Repository URL: https://github.com/ashley351/studentmanagementsystem

Team leader: Ashley Joanah Senkaali

1. Group members

NAME	STUDENT NUMBER	REG NUMBER	SIGN
Ashley Jonah Senkaali	2400703713	24/U/03713/PS	
Komuhimbo Prillar	2400706148	24/U/06148/PS	
Ssevume Richard	2400711375	24/U/11375/PS	
Naganze Mariam Kisakye	2400725160	24/U/25160/PS	
Kabali Ezra	2400704960	24/U/704960/PS	
Ssempijja Shorn	2400720186	24/U/20186/EVE	

Name	Github name	Contribution
Ashley	ashley351	Team leader, database design and relationships, merging and testing
Richard	RichardSsevume	Roles, permissions and seeders
Mariam	kisakyemariam22-crypto	Authentication, layout and navigation
Shorn	Shorn987	Student management, README and credentials
Prillar	prillarkomuhimbo-a11y	Course management, screenshots
Ezra		Course registration, role-based display, messages

2. Assigned tasks and branches

Name	Assigned tasks	Branch
Ashley	Q1: migrations, models, relationships, Eloquent queries; class template; user_id and lecturer_id links	feature/relationships, feature/template, features/add-links
Richard	Q2: seeders; Q3b and Q3c: roles and permissions; Spatie package	feature/roles-permissions-seeders
Mariam	Q3a, Q3d, Q3e: login, middleware, access protection; Q5a: layout; Q5d: navigation by permission	feature/authentication-layout
Shorn	Q4a: student CRUD; student views and validation; Q6a and Q6b: README and credentials	feature/student-management
Prillar	Q4b: course CRUD; course views and validation; Q6c: screenshots	feature/course-management
Ezra	Q4c: registration with duplicate prevention; Q4d: role-based display; Q4e: messages	feature/registration

3. Pull requests

All pull requests were submitted to `develop` and reviewed by the team leader before merging.

Contribution	collaborator	Branch	Status
Q1: Database design and relationships	Ashley	feature/relationships	Merged
Roles, permissions and Spatie install	Richard	feature/roles-permissions-seeders	Merged
Add class template layout and assets	Ashley	feature/template	Merged
Add roles, permissions and sample database seeders	Richard	feature/roles-permissions-seeders	Merged
Link students to users and courses to lecturers	Ashley	features/add-links	Merged
Add Breeze auth, role assignment and class template layout	Mariam	feature/authentication-layout	Merged
Add registration controller, routes and views	Ezra	feature/registration	Merged
Student CRUD	Shorn	feature/student-management	Merged
Course CRUD	Prillar	feature/course-management	Merged

4. Contributions

Ashley (team leader)

- Created the Laravel project and the GitHub repository with the `main` and `develop` branches, and invited the team.
- Designed the database: students, courses and course registrations tables, with foreign keys and a unique rule on student and course to prevent duplicate registrations.
- Wrote the models and the `hasMany belongs to` and `belongsToMany` relationships, and tested the Eloquent queries in Tinker.
- Added `user_id` on students and `lecturer_id` on courses, with the matching relationships.
- Added the class template (layout and assets) to the project.
- Reviewed and merged pull requests, resolved merge conflicts, and tested the integrated application.

Richard

- Installed and configured the Spatie permission package.
- Created the roles (administrator, lecturer, and student) and permissions.
- Wrote the seeders for sample users, roles, students, courses and registrations.

Mariam

- Set up authentication (login, logout and registration) with Laravel Breeze.
- Assigned roles on registration and protected routes with middleware.
- Built the layout and navigation using the class template.

Shorn

- Describe: student controller, routes, views, validation, README and credentials.

Prillar

- Describe: course controller, routes, views, validation, and screenshots.

Ezra

- Built the course registration controller, routes and views.
- Add: duplicate-registration check, role-based display, messages.

5. Testing and integration

The integrated application was tested on `develop` with the administrator, lecturer and student sample accounts before the final version was merged into `main`.

6. Evidence

The full commit history, branches and pull requests can be inspected on the repository

https://github.com/ashley351/studentmanagementsystem